An Energy Consumption Analytics Dashboard

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Abstract—This paper presents an end-to-end stateful digital-twin prototype for a simulated 7,500 m² commercial office. It connects telemetry ingestion, thermal-state estimation, driver forecasting, fault detection, scenario analysis, and comfort-constrained model predictive control (MPC) for HVAC and a 250 kWh battery. Real exogenous drivers comprise Berlin 2019 ERA5 weather, PVGIS-SARAH2 irradiance and PV output for a 250 kWp array, and German/Luxembourg day-ahead electricity prices from an Open Power System Data snapshot. Occupancy, indoor temperature, HVAC operation, lighting, plug loads, and the building are simulated; savings are therefore simulation results, not measured building outcomes. In the repository's 78-day test comparison, cost-focused MPC reduced cost from €5,868 to €5,440 (7.3%) and recorded zero comfort violations versus eight for the thermostat baseline, while HVAC energy increased from 21.81 to 26.24 MWh (+20.3%). The presentation reports a different energy increase (+33.2%), which requires reconciliation. Scenario analysis found €428 baseline savings, €57 with no battery, and €720 with doubled battery capacity. A flatline detector achieved 0.944 recall with four-hour latency on the injected stuck-temperature-sensor case. Findings are limited to one single-zone simulation.

Keywords—Building energy management, digital twin, model predictive control, fault detection, forecasting, battery storage, thermal comfort.

I. INTRODUCTION

Commercial buildings combine HVAC, lighting, plug loads, on-site generation, and storage. Energy demand responds to weather, occupancy, thermal inertia, equipment condition, and electricity prices. Forecasting and analytics can help, but prediction alone leaves the operator to connect information with control decisions. Reviews of building-energy prediction and control discuss the links among forecasts, comfort, optimization, and deployment [1]–[6].

This project builds a closed-loop prototype for a reference office. Hourly external drivers and simulated telemetry update a stateful building twin. The twin estimates its thermal parameters from observations, forecasts selected drivers, scores injected faults, evaluates counterfactual scenarios, and sends a 24-hour optimization problem to Pyomo with HiGHS. The first control action advances the simulated plant and produces the next telemetry update. The twin does not read simulator internals.

The work is a digital twin in architectural terms because information flows in both directions inside the simulation. It is not a twin of an occupied or metered real building. Its contribution is an integrated and testable simulation loop that reports cost, energy, peak demand, comfort, model bias, and fault-detection performance together.

II. LITERATURE SURVEY AND RESEARCH GAP

Amasyali and El-Gohary [1], Ahmad et al. [4], and Bourdeau et al. [6] review data-driven building-energy prediction. They describe weather and use patterns as important context for model performance. Building MPC research emphasizes forecasts, thermal models, comfort bounds, and control horizons. Killian and Kozek [2] identify practical questions for energy-efficient MPC; Drgoňa et al. [3] review formulations and deployment challenges; Oldewurtel et al. [7] study weather-aware climate control. Thermal identification research, including Bacher and Madsen [8], informs the project's telemetry-based RC model. Fault-detection reviews distinguish model-based, rule-based, and process-history methods [9], [10]. Isolation Forest provides a generic anomaly baseline [11]. Digital-twin studies emphasize synchronized data and bidirectional interaction [12], [13].

A. Ten-Study Comparison

Study	Method	Contribution	Limitation
Amasyali & El-Gohary, 2018 [1]	Prediction review	Maps energy predictors and drivers	No closed control loop
Killian & Kozek, 2016 [2]	Building MPC review	Frames benefits and practical issues	Depends on model and forecast quality
Drgoňa et al., 2020 [3]	MPC review	Unifies methods and application concerns	Integration remains challenging
Oldewurtel et al., 2012 [7]	Weather-aware MPC	Uses forecasts for climate control	Case-specific model and building
Bacher & Madsen, 2011 [8]	Thermal identification	Compares heat-response models	Quality depends on sensor signals
Katipamula & Brambley, 2005 [9],[10]	HVAC FDD reviews	Organizes methods and tradeoffs	Needs fault-aware validation
Liu et al., 2008 [11]	Isolation Forest	Generic anomaly isolation	Does not diagnose HVAC causes
Fuller et al., 2020 [12]	Digital-twin review	Enabling technology and open problems	Broad cross-domain scope
Boje et al., 2020 [13]	Construction digital twin	Semantic synchronized data flow	Construction rather than MPC focus
Afram & Janabi-Sharifi, 2014 [14]	HVAC control review	Surveys HVAC control strategies	Deployment varies by case

B. Research Gap

The literature often treats forecasting, fault detection, thermal identification, scenario analysis, or optimal control separately. This project connects these capabilities through a telemetry-mediated stateful loop and evaluates that loop on a reproducible simulated plant. The contribution is integration and evaluation, rather than a claim that an individual algorithm is new. It also measures how battery capacity, comfort-band width, and model bias affect outcomes.

III. OBJECTIVES AND CONTRIBUTIONS

The project objectives are to represent HVAC, lighting, plug loads, PV, battery, and grid interaction; ingest timestamped telemetry with quality metadata; identify thermal behavior without exposing simulator parameters to the twin; forecast external drivers; assess injected faults; compare thermostat and receding-horizon MPC; and provide scenario exploration in a local Streamlit dashboard. The contribution is a reproducible simulation architecture, a telemetry-only twin boundary, multi-objective reporting, and fault/scenario tests that expose limitations along with benefits.

IV. PROPOSED FRAMEWORK

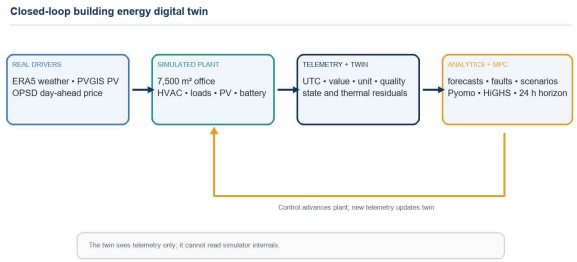


Fig. 1. Project architecture and closed-loop data and control flow.

A. Data and Reference Building

The processed driver file contains 8,758 hourly UTC rows in 2019. PVGIS-SARAH2 supplies plane-of-array irradiance and PVGIS calculates output for a 250 kWp array at 30° tilt in Berlin; ERA5 provides outdoor temperature and wind. Day-ahead prices come from the DE/LU series in the OPSD 2020-10-06 snapshot. Two hours without published prices are excluded rather than interpolated. The year includes 211 negative wholesale-price hours and prices from −€90.0/MWh to €121.5/MWh [15]–[18].

The reference office is a 7,500 m² single thermal zone with 250 kWp rooftop PV, a 250 kWh battery, a reversible heat pump, and weekday 07:00–19:00 occupancy. Occupancy, lighting, plug load, HVAC, and indoor temperature are simulated. This is a hybrid dataset: real external drivers and a simulated building interior. Calibration checks compare modeled quantities with published office ranges but do not make the plant empirical.

B. Processing and Evaluation Split

PVGIS timestamps are floored to the hour before joining OPSD prices. UTC is canonical; local time is derived for schedules and solar position. Forecast evaluation uses a chronological split: 5,832 training hours through September 1, 1,056 validation hours through October 15, and 1,870 test hours from October 15 through December 31. Random shuffling is avoided to reduce future-to-past leakage. The headline MPC comparison uses the repository's 78-day test window.

C. Stateful Twin and Thermal Identification

Telemetry records carry timestamp, value, unit, and quality. Twin-state logic handles validity, staleness, ordering, and missing readings. A resistance-capacitance thermal model estimates UA and effective thermal capacitance from observations. The twin receives sensor values and site-survey entities; a source test enforces that twin modules do not import the plant simulator. Identified UA is 5.83 kW/K versus simulator value 5.00 (+17%); capacitance is 357 versus 250 kWh/K (+43%); and time constant is 61 versus 50 h (+22%).

D. Driver Forecasting and Fault Detection

Persistence, seasonal mean, Ridge, and gradient boosting are compared on 1,032 evaluation samples. Ridge is selected for outdoor temperature and base load; seasonal mean is selected for irradiance. Model selection uses error and skill against persistence. Forecasts are point estimates, so the MPC is not currently uncertainty-aware.

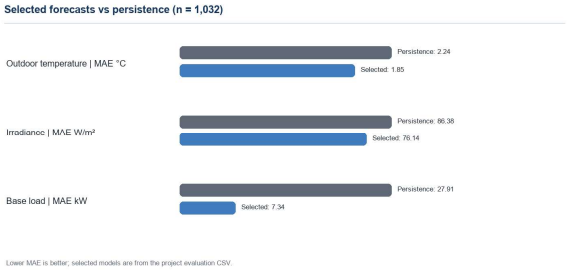


Fig. 2. Mean absolute error for persistence and selected forecast models.

Fault tests inject HVAC-efficiency decay, PV soiling, a stuck temperature sensor, a failed HVAC stage, and a drifting meter at known times. Residual thresholds and rolling statistics flag deviations; a flatline detector looks for a lack of sensor variation. A frozen temperature sensor can cause both the previous observation and next predicted state to remain nearly unchanged, leaving a small residual. For the injected stuck-sensor case, residual-based detectors had zero recall; flatline detection reached 0.944 recall at four-hour latency.



Fig. 3. Stuck-sensor fault recall; the result is specific to the injected case.

E. Scenario Engine and MPC

The scenario engine changes inputs without mutating baseline data. Scenarios include occupancy +20%, doubled PV, doubled battery, no battery, HVAC efficiency −25%, HVAC capacity −50%, prices +50%, and a ±1 K comfort band. MPC formulates HVAC and battery decisions as a mixed-integer optimization using Pyomo and HiGHS. It optimizes over 24 hours, applies the first action, advances the plant, and repeats. Comfort bounds and HVAC mode constraints encode operating limits. Cost-focused, cost-plus-energy, and cost-plus-peak variants are compared.

The reference delivered tariff adds €60/MWh to wholesale prices. At this tariff, the project reports three hourly cases in 2019 where the LP relaxation permits simultaneous heating and cooling; the binary mode constraint is retained for physical correctness. The larger count in an initial planning probe applied raw wholesale prices and is not the reference tariff result.

V. IMPLEMENTATION AND PROJECT PLAN

Implemented modules include driver ingestion, the simulated plant, twin state, thermal identification, forecasting, fault injection and scoring, scenario analysis, optimization/MPC, and a local Streamlit dashboard. DataProvider implementations support CSV replay and a synthetic stream. FastAPI, MQTT, database services, and Docker are described as deferred. The presentation identifies timestamp alignment, state update, scenario isolation, and stuck-sensor checks; it states that a full end-to-end test is specified but does not report a full test-suite pass total.

Phase	Work	Output
1. Scope and data	Define office, assets, UTC sources, simulation boundary	Processed driver data and assumptions
2. Plant and telemetry	Implement RC building, loads, PV, battery, quality-tagged stream	Simulator and replay/synthetic providers

Phase	Work	Output
3. Twin and analytics	Maintain state, identify thermal model, forecast and score faults	Twin, forecasts, residual and flatline detectors
4. Control and scenarios	Formulate comfort-constrained MPC and counterfactual runs	Pyomo/HiGHS controller and scenario engine
5. Evaluation and UI	Compare baseline, inspect sensitivities, expose dashboard	78-day results, scenario outputs, local Streamlit UI

VI. EXPERIMENTAL RESULTS

A. Forecast Selection

Target	Selected model	MAE	RMSE	R^2	Skill vs persistence
Outdoor temperature	Ridge	1.847 °C	2.399 °C	0.709	+17.5%
Irradiance	Seasonal mean	76.137 W/m²	145.734 W/m²	0.597	+11.9%
Base load	Ridge	7.338 kW	12.348 kW	0.963	+73.7%

These are model-selection outputs on 1,032 samples, not guarantees of future-year accuracy. Skill is relative to persistence. Irradiance errors are measured in W/m² and are not directly comparable with temperature or load errors.

B. Cost, Energy, Peak, and Comfort

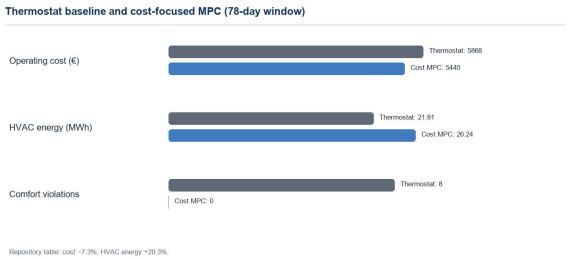


Fig. 4. Repository results table for the 78-day controller test.

Controller	Cost	HVAC MWh	Peak kW	Violations
Thermostat	€5,868	21.81	337	8
Thermostat + greedy battery	€5,864	21.81	337	8
MPC, cost	€5,440	26.24	400	0
MPC, cost + energy	€5,435	24.87	364	0
MPC, cost + peak	€5,668	27.46	220	0

The repository table shows about €428 (7.3%) lower operating cost under cost-focused MPC and no recorded comfort violations, compared with eight under the thermostat. HVAC electricity rises 4.43 MWh (+20.3%) and peak rises to 400 kW from 337 kW. Cost-plus-energy reduces HVAC electricity relative to cost-only MPC; cost-plus-peak lowers peak demand to 220 kW but costs more than cost-only MPC. The objectives are not interchangeable.

Source discrepancy: the slide deck and decision log state +33.2% HVAC energy and −27.1% HVAC cost for cost-focused MPC. The README's underlying 78-day table reports 21.81 versus 26.24 MWh (+20.3%) and €5,868 versus €5,440 (−7.3%). This paper uses the repository table as its primary result; the slide and decision-log figures need reconciliation against the original run before publication.

C. Scenario Sensitivity

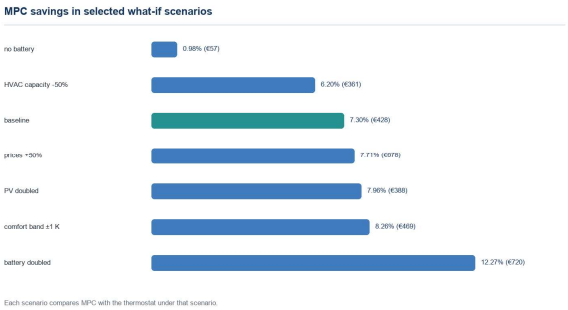


Fig. 5. Selected scenario savings relative to each scenario thermostat baseline.

Scenario	Saving	Rate / note
Baseline	€428	7.30%
No battery	€57	0.98%
Battery doubled	€720	12.27%
PV doubled	€388	7.96%
Comfort band ±1 K	€469	8.26%; HVAC 20.52 MWh
HVAC capacity −50%	€361	6.20%; 13 MPC / 44 baseline violations

Removing the battery reduces savings from €428 to €57, while doubling capacity raises savings to €720. The repository attributes roughly 87% of baseline value to the battery and 13% to thermal mass. A wider ±1 K comfort band is the only tested lever reported to improve both cost and energy: savings rise to €469 and HVAC energy falls to 20.52 MWh. These results are specific to this scenario model, tariff, and test period.

D. Model Bias and Fault Findings

The repository reports that MPC with true plant parameters saved 7.9% but produced 269 comfort violations, while MPC with the biased identified model saved 7.3% and produced zero. The project attributes this to a conservative operating margin under variable loads; it does not establish that inaccurate models are generally safer. Residual methods miss the frozen sensor when prediction and observation freeze together. A flatline rule achieved 0.944 recall with four-hour latency on the injected case. A drifting meter and degraded HVAC can produce similar HVAC residuals, so independent measurements are needed for reliable diagnosis.

VII. DISCUSSION

The integrated result is more informative than a single efficiency score. Cost-focused MPC shifts operation toward lower-price hours, but preheating and precooling widen temperature excursions and increase modeled envelope losses. Energy-aware and peak-aware objectives alter the tradeoff. Scenario analysis also challenges the initial expectation that building thermal mass would dominate flexibility: battery capacity has a strong effect in this configuration, while shifting energy through the building fabric is bounded by comfort and may add heat loss.

The prototype supports exploration and methodological demonstration, not a claim of real-building savings. Simulated interior loads, one thermal zone, constant COP, no free cooling, a daily demand-charge proxy, point forecasts, one year of weather and prices, and a 78-day headline control test limit interpretation. Field validation requires measured telemetry, commissioning, and safe supervisory control.

VIII. LIMITATIONS AND FUTURE WORK

Key limitations are the absence of a real building connection; a single-zone thermal representation; constant heat-pump COP and no economizer, window, or night-purge cooling; a daily demand-charge proxy instead of a monthly tariff; point forecasts without

uncertainty propagation; and injected faults whose labels may not reflect field prevalence. Future work should integrate measured multi-zone telemetry, quantify forecast uncertainty and comfort risk, evaluate longer seasonal periods and real tariff structures, improve sensor-fault diagnosis with independent measurements, and add secured gateways before any field trial.

IX. CONCLUSION

The project implements a stateful energy digital-twin prototype that closes a telemetry-to-control loop for one simulated commercial office. Repository results show €428 (7.3%) lower cost under cost-focused MPC with zero comfort violations, alongside a 20.3% increase in HVAC electricity in the results table. Battery sizing strongly changes modeled savings; flatline detection addresses a stuck-sensor case missed by residual methods. These findings are simulation-specific. Reconciling the presentation's HVAC-energy figure and validating on measured building data are required before making real-world performance claims.

X. DECLARATIONS

Data and software availability—The project repository and data provenance are listed below. Weather and PV inputs originate from European Commission JRC PVGIS; the price series is from Open Power System Data. Interior building signals are simulated. The work is a simulation study and reports no real-building deployment or measured savings.

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